The Racialized Packaging of Punishment: An Instrumental Variables Approach to the Cerberus of Punishment

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Abstract

Research on racial disparities in the criminal justice system has generally focused on isolated sentencing decisions, with less attention to the broader "package" of punishment - the mix of incarceration, probation, monetary sanctions - that a defendant may receive. Using court administrative data in Minnesota from 2010-2018, we use multivariate regression and instrumental variables models to simultaneously estimate three different modes of racialized punishment: control of the body (incarceration), more generalized surveillance (probation), and financial extraction (monetary sanctions). We instrument incarceration length using local jail capacity, which accounts for confounding and the endogenous simultaneity of confinement and the other forms of punishment. Our results indicate that the "mix" of punishment received for similarly situated defendants is patterned by race. Multivariate models suggest that Black, Hispanic, and Native American defendants are receiving lower fines and fees but higher confinement and surveillance loads relative to White defendants. After incarceration is taken into account, however, Black, Hispanic, and Native American defendants receive *higher* monetary sanctions relative to White defendants. This suggests that the lesser monetary punishment, for some groups, is due to a "trade off" with confinement. These results show that punishment can, and perhaps, should be modeled as it is often experienced -- as a constitutive package of costs, surveillance, and confinement.

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Despite decades of research on the relationship between race and criminal sentencing, standard practice has been to measure forms of punishment in isolation, that is, estimating separate models for incarceration, probation, monetary sanctions, and other criminal sanctions. Yet among those who receive them, these punishments are experienced in combination as what we term a "package of punishment," or the combination of punishment modes and intensities that constitute a criminal sentence. Little is known about how race impacts the overall package of punishment, nor how race may differentially shape varied aspects of the punishment package.

This paper develops a more comprehensive account of the role of race and ethnicity in criminal sentences involving a more complete package of punishment, including incarceration, probation, and monetary sanctions, each corresponding to a distinctive *axis* of racialized social control. In doing so, we make both a theoretical and methodological contribution. Punishment, as it is often experienced, is often not experienced with just an isolated form of punishment and its downstream consequences. Rather, it is a matrix of costs, confinement, and surveillance that are experienced, and structure individuals' choices and actions, *in tandem.* This "cerberus" - making reference to the mythical three-headed dog - of punishment is more than just the additive sum of its parts: rather, each mode of punishment may interact with the others in terms of impacting individual defendants, exacerbating financial and familial strain (Harris et. al. 2010; Comfort 2009).

Methodologically, modeling these outcomes separately likely biases estimates of residual race effects. Different forms of punishment may be racialized in different ways, and may be

correlated with each other in the sentencing process. For example, a judge who gives a more extended incarceration sentence may reduce the fine amount they order. Put another way, we may make erroneous conclusions about how a certain punishment axis is racialized if we do not consider that components interrelationship to other modes of punishment, which, in turn, *are also raced.* In effect, we risk not only missing part of the picture of punishment a defendant receives by only considering each component in isolation, the analysis also may yield a distorted picture of the sentencing process. Therefore, we also model this punishment process as it is often experienced: as a simultaneous "package of punishment" that incorporates multiple axes of punishment. Further, we leverage an instrumental variable design to explore the interrelationships between the modalities of punishment, which allows a more resolute picture of the remaining racial sentencing disparities. In doing so, we draw on a novel analysis of court administrative data in Minnesota, a state characterized by relatively low rates of incarceration, relatively high racial disparities, and lengthy probation terms (The Sentencing Project 2021). In presenting a more complete and accurate picture of racial disparities in the sentencing process across multiple modalities of punishment, we gain further insight into how racial stratification is (re)produced within the sentencing stages of the criminal legal system.

Punishment and Racialized Social Control

Criminal punishment is strongly racialized in the United States. The steep rise in jail and prison populations from the mid-1970s to 2008 is the story of racialized mass incarceration, with Black men experiencing especially high rates of imprisonment and felony conviction (Western 2004; Shannon et al. 2017). Researchers have attributed such disparities to diverse causes, including overt discrimination, color-blind racism, political fear-mongering, criminalization in

response to racial threat, rising violence in impoverished communities, structural economic shifts, or some combination of these mechanisms (Alexander 2012; Campbell and Schoenfeld 2013; Forman 2017; Garland 2001; Tonry 1995, 2009, 2011; Wacquant 2000, 2001).

Although the imprisonment rate has dropped over the last decade (≈17% from 2009-2019), considerable racial disproportionality remains. According to the Bureau of Justice Statistics (BJS), in 2019 33% of the prison population was Black, 23% was Hispanic, and 31% was non-Hispanic White (Carson 2020). The demography of imprisonment in the United States, in conjunction with its sharp and unparalleled rise, are the two components of *mass imprisonment* (Garland 2001) and *racialized mass incarceration* (Bobo and Thompson 2006). Accordingly, racial disparities in incarceration are well documented and are powerfully illustrated through simple comparisons. In state prisons, Latinos are incarcerated at 1.4x and African Americans are incarcerated at 5.1x the rate of Whites (Nellis 2016). These disparities are even more pronounced when men are examined separately; Latino men are 2.5x and African American men are 5.7x more likely to be incarcerated than White men (Carson 2020). Racial disparities are also prevalent in the population of those with felony criminal records who are no longer under any form of correctional supervision. Shannon et. al. (2017) estimate that 8% of Americans as of 2010 had a felony conviction, and 33% of African American men held a felony record. The aggregate punishment disparities described here "can be located at a variety of decision-making points (Baumer 2013, p.236)", including both selection into arrest as well as post-arrest racial biases in criminal justice processing. Research indicates that only about 55% of the racial disparity in incarceration rates is due to racial variation in the arrest stage, leaving a substantial portion of the disparity due to post-arrest factors such as racial variation in sentencing (Baumer 2013; Tonry and Melewski 2008).

These racialized patterns in punishment have been traced to the coupling of criminal justice and poverty governance, where rollbacks - or redirections - of the social assistance sector coincided with the immense growth of the criminal justice apparatus (Wacquant 2009; Soss et. al. 2011). In this theoretical framework, the prison serves not as a deterrent and incapacitative crime control device, but rather as an institution "for the capture and management of a population considered contemptible and expendable (Wacquant 2009, p. 195)," in lieu of declining social welfare. Wacquant (2000) identifies the ethnoracial character of modern U.S. punishment as a "surrogate ghetto," where the prison has emerged as the preeminent social control device for poor, Black Americans that mirrors the function of city blocks segregated by race. This characterization of penal institutions is a later development in racialized forms of social control that included chattel slavery and the Jim Crow system of enforced discrimination. This literature identifies punishment, and, in particular, the role of incarceration and the prison, as a defining feature of social control and racial oppression in American society founded upon the criminalization of Black and minority populations (Hinton and Cook 2021).

Further, scholars have noted how both crime and punishment have been racially typified in American culture and practice. Support for punitive policies are grounded in a "belief system that constructs crime in terms of race and race in terms of crime (Beckett and Sasson 2000, p. 136)." The disparate rates of arrest and incarceration have been used as part of a "statistical discourse" (Muhammad 2010) that has, in part, helped create stereotypes about Blackness and crime in American culture and public opinion. A wealth of scholarship indicates the coupling of Blackness with criminality (e.g., Quillian and Pager 2001), and this nexus helps explain patterns in punitiveness. Using national survey data, Chiricos et. al. (2004) show that racial typification of crime is a robust predictor of punitiveness, and this could extend into racialized forms of

punitiveness as well. A robust literature also marks individual-level punishment as shaped by race, of which following sections will detail the racial effects on punishment across three different, but interrelated, modes: incarceration, probation, and monetary sanctions.

Race and Incarceration

In many ways, racialized mass incarceration has been a product of changes and enduring inequities in punishment. This body of scholarship on race and imprisonment explores race as an individual-level extralegal variable that leads to disparate outcomes in sentencing (Baumer 2013; Spohn 2000). Early scholarship in this area tended to measure sentencing severity with either a single variable of sentence length (indicating the duration of the imposed sentence), a single binary variable (coded 1 if an individual was sentenced to any incarceration and 0 for any other sentence such as probation or monetary penalties), or a combination of both measures. Research in this tradition typically controls for legal variables such as offense severity and criminal history, as well as other case or individual characteristics (e.g., gender) and examines the residual racial effects that remain.

Generally, studies examining the binary measure of incarceration find that racial and ethnic minorities are not only more likely to be convicted, but also receive harsher sentences than White defendants (Britt 2000; Chiricos and Crawford 1995; Spohn and Holleran 2000; Steffensmeier and Demuth 2000; Unnever 1982; Unnever and Hembroff 1988). Analyses of sentence length and those that combine both measures of sentence severity yield less consistent results, but often conclude that at least in certain contexts (Peterson and Hagan 1984; Ulmer and Johnson 2004) or among specific subgroups (Demuth and Steffensmeier 2004; Lizotte 1978) racial minority defendants are punished more harshly than White defendants. Further, these

extralegal racial effects exhibit significant variation across counties, and punishment is also shaped by county-level attributes, including racial demographic characteristics (Ulmer and Johnson 2004).

Much of the scholarship in this arena explores only a few racial/ethnic groups (White, Black, and Hispanic). Among these categories, Black defendants are treated the most punitively, followed by Hispanic defendants, with White defendants the least punitively (Frase 2009; Unnever 1982; Unnever and Hembroff 1988). When age and employment are considered, research has further indicated that men of color who are younger or with worse employment circumstances face more severe incarceration sentences (Lizotte 1978; Spohn and Holleran 2000; Steffensmeier and Demuth 2000). Importantly, recent scholarship documents that, at least in US Federal Court, race has played a declining role in patterning incarceration lengths, which has been driven by both White increases and Black decreases in sentence lengths (Light 2021). Consistent with this extensive body of research, our first research hypothesis is as follows:

> *Hypothesis 1*: Black, Hispanic, and Native American defendants will receive longer incarceration sentences, net of other observed variables.

Race and Probation

While less prolific, another body of scholarship has investigated how race impacts an individual's probation sentence, typically finding that racial minorities receive harsher or longer sentences. Most of this literature has focused on the likelihood of receiving probation versus a custodial sentence. For instance, studies by Freiburger and Hiliniski (2013) and Harrington and Spohn (2007) found that Black defendants were more likely to receive a sentence of jail rather

than probation. Likewise, Black men are the least likely to receive probation instead of incarceration relative to whites and women (Freiburger and Sheeran 2020). Among those receiving probation sentences, Lowder et al. (2019) found that among those assigned to a low-risk category by an assessment, white defendants received shorter probation lengths than black defendants. Research also indicates that disparities in probation supervision rates are smaller than those for incarceration rates (Phelps 2018).

Given the extant findings between race and probation sentencing, coupled with the scholarship on incarceration as a particularly racialized form of social control (e.g., Wacquant 2009), this leads us to an additional research hypotheses:

> *Hypothesis 2:* Black, Hispanic, and Native American defendants will receive longer probation sentences, net of other observed variables.

Race and Monetary Sanctions

Recent research on criminal punishment in the United States has emphasized monetary sanctions as well as incarceration and community supervision. Monetary sanctions are financial penalties imposed by the state in criminal cases, such as fines, fees, surcharges, and restitution. In a series of articles and a recent book, Alexes Harris (2016) investigates the relationship between race/ethnicity, context, and monetary sanctions. Harris, Evans, and Beckett (2011) find that Hispanic defendants receive higher fines and fees on average as compared to White defendants in an analysis of Washington data. They also find that racialized crime scripts, tied to both moral outrage and cultural stereotypes, impact the amount of monetary sanctions. For example, Black defendants charged with violent crimes are levied higher fines and fees than

Black defendants charged with non-violent crimes. Moreover, they find that Latinx defendants receive higher fines, especially in the context of counties with large Latinx populations. In such contexts, even persons who are not Latinx receive racialized "courtesy stigma" (Goffman 1963) resulting in high fines for drug crimes. Similarly, persons convicted of violent crimes in counties with greater proportions of black residents also receive higher monetary sanctions. This relatively new body of research highlights the salience of race in sentencing decisions *beyond* confinement or community supervision, and motivates another research hypothesis:

> *Hypothesis 3:* Black, Native American, and Hispanic defendants will receive larger monetary sanction orders, net of other observed variables.

Linking Incarceration and Other Forms of Punishment

Importantly, little research into the racialization of punishment has considered multiple types of sanctions simultaneously in their analyses. In other words, studies have primarily analyzed each mode of punishment in isolation, and have not examined the interrelationships between different axes of punishment, nor the potential heterogeneity in race effects across different sentencing outcomes. A related exception is Kutateladze et. al. (2014), who examine the racial effects on multiple different decision-making points, including pre-trial detention, plea offers, and incarceration. They found that Black and Latino defendants were more likely to be detained, offered pleas, and to be incarcerated, but were also more likely to receive case dismissals. This scholarship illustrates that different aspects of the criminal justice apparatus can be racialized in different ways and to varying degrees. While the goal of the current paper is to assess the racial impacts of different axes *within the sentence*, rather than different decision

points across the criminal justice process, a related logic applies: that distinct aspects of the criminal justice system are differentially racialized and interrelated to an extent. Thus, more complex analyses are needed to address the disparities within. Key to this idea is the interrelationships between axes of sentencing, particularly in terms of the main outcomes of the criminal sentence: confinement, probation, and monetary sanctions. Most defendants who receive a sentence of incarceration also receive some form of monetary sanction (Beckett and Harris 2011; Harris 2016; Harris et al. 2017; Harris, Evans, and Beckett 2010). Do fines constitute a substitute for incarceration length? Or is the relationship between the two positive, where defendants are given higher fines and fees that accompany longer incarceration spells?

In an early study of fraud defendants in the federal courts, Waldfogel (1995) finds evidence of substitution, observing a negative association between monetary penalties and the likelihood of prison and the severity of prison sentences. A more recent study of federal white-collar sentencing similarly finds that defendants who pay fines are likely to receive less prison time, but restitution has a positive, albeit diminishing, impact on incarceration length (Schanzenbach & Yager 2006). In a more recent and comprehensive analysis of federal sentencing data, Martin (2012) finds evidence that monetary sanctions function as both a complement and substitute to incarceration. Specifically, Martin finds that receiving a prison sentence is associated with reduced monetary penalties, suggestive of a "trade-off", but that prison lengths are associated with higher monetary loads. To date, we are not aware of any studies that estimate the impact of one component of punishment on another form while addressing the simultaneity bias that arises. Further, we are not aware of any such study examining monetary sanctions, probation, and incarceration using state court data among the much broader class of crimes for which both fines and incarceration may be imposed. Although

the research in this area is in its nascent stages, the preponderance of the evidence is suggestive of a trade-off of incarceration and lower levels of punishment. Thus, our fourth and final hypothesis is:

Hypothesis 4: Incarceration will be negatively associated with both probation length and monetary sanction orders.

Current Study

The Racialized Packaging of Punishment

The current study aims to estimate the extralegal race effects on three separate modes of punishment: incarceration, probation and monetary sanctions, and interrogate the differences in racialization that may exist across these axes of punishment. This study extends the research on the extralegal effects of race on punishment in two important ways. Baumer (2013) identifies narrow racial categorizations as a significant limitation in the body of regression-based study of sentencing outcomes, which focuses primarily on Black-White disparities. This study uses a more expansive racial and ethnic categorization than previous work, with sufficient sample sizes to adequately estimate the punishment patterns for smaller racial groups, such as Indigenous Americans.

Second, we examine a broad class of state court data to simultaneously model incarceration, probation, and monetary sanction outcomes, and examine the similarities and differences in how individual-level case and defendant characteristics impact each axis of sentencing. We expect that certain case or defendant characteristics may weigh more heavily in particular parts of the punishment process, especially regarding the disproportionate incarceration of people of color (e.g., Wacquant 2009). Thus, we risk missing part of the picture

of punishment a defendant receives by only considering each component in isolation. Further, this allows us to examine how the racialization of punishment may vary with the mode punishment under consideration.

Correlations Within the Punishment Matrix

From a statistical perspective, studies that examine only one of the punishment components risk model misspecification if one component influences another aspect of the punishment defendants receive. To the extent that punishment outcomes influence one another, and are correlated with key exposure variables of interest, leaving other forms of punishment out of the model could bias the estimated effects. Our final contribution is to model the effect of the incarceration component on both the monetary sanction and probation components to alleviate the bias that could result from key individual extralegal variables being collinear with a neglected sentencing component. This allows us to a) investigate whether a judge's decisions represent a “piling on” (Uggen and Stewart 2014) or a “substitution” of punishment forms, and b) control for the interrelationships between race and incarceration, to obtain a more accurate picture of the extralegal effects on monetary sanctions and probation.

While we cannot disentangle the causal ordering of these sentences with our current data, previous research indicates that incarceration is typically considered the most severe of these penalties by court actors. For instance, in a developed scale of sentencing severity derived from the opinions of judges imprisonment was considered more severe than probation—even when the duration of probation was far lengthier than the prison term and monetary sanctions were not even considered in this scale (Buchner 1979). Clair (2020) more recently confirmed that courtroom actors - even public defense attorneys- consider confinement paramount even though some defendants disagree with this assumption. Thus, previous research suggests that

incarceration decisions are of prime interest to the involved parties, and may represent a key decision point in the sentence in which other sanctions, such as fines, are endogenous to. We therefore control for the impact of incarceration using an instrumental variable design described below. With these advances, the impact of racial characteristics on multiple aspects of punishment will come into sharper focus.

Data and Measures

Our analysis draws on an extract of administrative criminal court data from the Minnesota State Court Administrator's Office (SCAO). The SCAO data extract includes granular data points for all cases filed from 2010-2015 in Minnesota criminal courts. Data points include charge-level and case-level details and events as well as defendant demographics and residential information. We restrict our analyses to misdemeanor, gross misdemeanor, and felony level cases. While petty misdemeanors represent a significant aspect of the criminal legal system, particularly in the imposition of fines and fees (Harris et. al 2017), they often do not entail any incarceration or probation terms, and therefore are not comparable to more serious cases. Our first key endogenous variable, total fine and fee order, is composed of all fines and fees levied against a defendant following sentencing. Detailed ledger information on court fines and fees - otherwise known as legal financial obligations (LFOs) - is compiled from two data storage systems, the Minnesota Court Information System (MNCIS) and the Violations Bureau Electronic System (ViBES). Minnesota's two largest counties by population, Hennepin and Ramsey, have incomplete ViBES data due to a system transition pre-2010, so we analyze the period that includes the most complete and comparable data. We adjust all monetary amounts to January 2018 dollars to adjust for inflation.

Our second endogenous variable, ordered incarceration length, is measured as actual days in prison and/or jail sentenced, minus any jail credit, stayed time, or supervised release periods. Probation days consist of the probation sentence, expressed in days. We also use the SCAO data to construct indicators of whether the case went to trial, and a proxy of criminal history that measures the number of prior cases a defendant has within the data extract. Martin (2012) states that models "that exclusively focus on likelihood of each sanction neglect the importance of severity at the risk of mischaracterizing" the relationships among various sentencing outcomes. We therefore model each endogenous variable as a continuous amount.

To measure individual defendant characteristics, we use SCAO party information. Our focal variable in the analysis is the self-identified racial identity of the defendant as provided to the court or through a presentence investigation interview. We collapsed the SCAO racial categories into six groups: White, Black, Hispanic, Asian/Pacific Islander, Native American, and Other Race (which includes "Other," "Multiracial," "Unavailable," "Unknown," and "Refused"). Minnesota maintains a "state payables list," or a list of petty misdemeanor and some misdemeanor offenses with defined LFO amounts that are payable without an appearance in court. Therefore, lower offense levels have substantial missing defendant data (e.g., race), whereas more serious charges have more complete information on defendant characteristics. Because race is collected at the case-level and not the person-level, within-person racial self-identification cannot be assumed to be static but instead can change over time (Liebler et al. 2017; Liebler, Bhaskar and Porter 2016; Liebler and Ortyl 2014). To partially remedy this, we impute the modal race and gender category to all cases within defendants, which gains us purchase on defendant characteristics for lower level cases for which there is a higher level case with the information for that individual. Further, we include an explicit control for missing race

to adjust the estimates for the presence of missing race data. We also include age and a binary indicator for gender. Other legal and case characteristics are constructed from the SCAO charging information. We include legal controls of the number of prior convictions to the current case, the amount of time served credited by the judge, and an indicator of whether the case went to trial. We include a public defender variable indicating whether a public defender ever appeared at any point in the case as a proxy for ability to pay. We also control for the level of the highest convicted charge (felony, gross misdemeanor, or misdemeanor) in each case. Finally, we construct a measure of offense type by categorizing the charge into four different categories: violent, drug, alcohol/DUI, and other offenses.

Our instrumental variable, the jail capacity ratio, comes from The Vera Institute's Incarceration Trends Dataset (Vera, 2020), which is merged at the county-year level. We calculate jail capacity ratios at the county and year level by taking the average daily number of confined persons in the jail and dividing by the jail-rated capacity. Therefore, values above 1 indicate jail overcrowding, whereas values under 1 indicate jail vacancies relative to stated capacity limits. This instrumental variable is described in further detail in the Analytical Strategy section below. Descriptive statistics for all variables in the SCAO data is included in Table 1. After listwise deletion of missing values, this leaves us with : 1,371,751 criminal cases.

Table 1: Case-Level Descriptive Statistics for SCAO Data

Variable	Mean	SD	Min.	Max.
Endogenous Measures				
Total Fine/Fee Order	302.78	2901.85	0	356515
Incarceration Days	25.78	297.08	0	41977
Probation Days	235.27	576.57	0	133225
Exogenous Measures				
White	.45	.50	0	1
Black	.17	.37	0	1
Hispanic	.06	.25	0	1
Asian	.03	.16	0	1
Native American	.04	.19	0	1
Other Race	.01	.12	0	1
Missing Race	.24	.43	0	1
Male	.71	.46	0	1
Age	33.00	12.40	15	115
Trial	.01	.01	0	1
Priors	6.26	8.36	0	236
Public Defender	.22	.41	0	1
Days Credit	6.04	117.76	0	129564
Felony	.06	.24	0	1
Gross Misdemeanor	.09	.28	0	1
Misdemeanor	.85	.35	0	1
Violent	.05	.22	0	1
Drug	.03	.17	0	1
Alcohol/DUI	.15	.36	0	1
Other Offense	.75	.43	0	1
Instrumental Variables				
County-Level Capacity Ratio	.75	.15	.2	1.15

Analytic Strategy

Our research design is threefold. It begins with a presentation of the endogenous variables - confinement length, probation length, and monetary sanction amount - by race. We additionally adjust the fine/fee orders by the inverse ratio of the race-specific median income to the overall median income[2], to examine the fine/fee orders relative to the existing average racial disparities in ability to pay.

[2] These data come from the 2018 ACS 5-year estimates of median income.

The second component is to estimate a multivariate linear regression model, treating both total fine/fee order, incarceration days, and probation days as endogenous variables. This method simultaneously estimates the effects on multiple dependent, or endogenous, variables. While the estimated coefficients and standard errors are identical to those when the models are estimated separately, the multivariate method takes into account the covariance of predictors across models, as errors in one model may be correlated to errors in another. The model is parameterized as follows,

$$Y_{ik} = a_{0k} + \sum \beta_{jk} X_{ij} + e_{ik}$$

where Y_{ik} is the *k*-th response for the *i*-th observation. β_{jk} is the *j*-th predictor's regression slope for the *k*-th response. a_{0k} is the regression intercept for the *k*-th response. X_{ij} represents the *j*-th case or defendant level characteristics for the *i*-th observation. e_{ik} are the multivariate Gaussian errors. The covariance structure is a 3x3 matrix with error variances on the diagonals and error covariances on the off-diagonals. We use the natural logarithm of all endogenous variables to reduce skewness in their distributions.

The third and final component of our analysis is to estimate models of total fine/fee order and probation length including the incarceration decision as an independent predictor. If one's LFO or probation amounts are determined, in part, by the incarceration length, then the above models will be misspecified. For example, if race is both a predictor of fines/fees and incarceration, and incarceration has an independent impact on fines/fees, then the race coefficients will be biased as incarceration confounds the race-LFO relationship.

Estimating the effect of incarceration length on fines/fees and probation is complicated by two sources of bias. First, if judges adjust their confinement length on the basis of fines/fees

or probation sentences, or vice-versa, we are confronted with a circular causal chain in which it is difficult to pin down the direction of the causal arrow, also known as simultaneity bias. If fines/fee orders are determined in this way, it poses an endogenous simultaneity bias problem that interferes with the identification of the impact of incarceration length on monetary sanctioning. The second source of bias is the traditional case of an omitted variable. In this case, an unobserved factor that causes both changes in the sentencing of incarceration length and the sentencing of another punishment component may lead to the wrong conclusion that the former caused the latter. One example of an omitted variable would be a judge's general level of punitiveness. Punitive judges may sentence more punitively on both components, thereby inducing a correlation, although a causal link between the two does not exist.

To deal with these endogeneity concerns, we develop an instrumental variable approach. Instrumental variables estimation allows for consistent estimation of causal effects in the presence of endogeneity by statistically isolating the part of the endogenous predictor that is uncorrelated with the error term. For example, Levitt (2002) instrumented the size of the police force with the number of municipal firefighters per capita to estimate the effect of police size on crime rates. Similarly, Sharkey et. al. (2017) estimate the causal effect of community nonprofits on neighborhood crime by instrumenting the change in community nonprofits with the change in nonprofits focusing on the arts, medical research, and the environment.

We use the county-level time-varying[3] change in jail capacity ratios as an instrument for confinement length. This model includes an important identifying restriction. It specifies that changes in jail population relative to capacity will be associated with incarceration length sentencing decisions through perceptions of overcrowding or the availability to fill vacancies, but that the capacity ratio will have no direct impact on monetary sanctions or probation apart

[3] This variable is at the county-level, not the individual-level.

from its indirect path through confinement length, net of other observed variables. To serve as a valid instrument, the capacity ratio must meet three conditions or assumptions. First, the relevance condition requires that the instrument must induce a change in the endogenous independent variable, incarceration length, and that this change will translate into a sufficiently strong relationship between them. Second, the exogeneity assumption requires that the change in capacity ratios should be uncorrelated with previous trends in the outcome variables. In other words, the values of the instrument should be independent of previous patterns in monetary sanction and probation sanctioning (i.e., should be distributed as though they are random), conditional on other variables in the model. Third, the exclusion restriction states that the instrument can only affect monetary sanctions through its impact on confinement length.

We show that there is a strong relationship between jail capacity ratios and sentenced confinement length, which gives direct evidence on the validity of the relevance condition. We follow the convention in the literature and consider the instrument to be sufficiently strong if the F-statistic from a Wald test on the instrumental variable is larger than 10 in the first stage regression (Stock and Yogo 2005). As Table 3 shows, the F-test on the instrument in the first-stage regressions yields a statistically significant F-statistic well above this value. We also examine whether lagged monetary sanction and probation patterns explain changes in the jail capacity ratio. We find no evidence that the changes in the jail capacity ratio are correlated with prior trends in monetary sanction sentencing or probation sentencing[4]. Finally, we argue that the instrument meets the exclusion restriction because monetary sanctions and probation lengths are unlikely to be directly impacted by jail capacities, apart from any perceptions or concerns with capacity that impact sentenced amounts through confinement. If the instrumental variable assumptions are satisfied, it is possible to estimate the causal effect of confinement length on the

[4] Models available by request to the author, will be included in an appendix at further date.

other aspects of the punishment package. This approach yields an estimate of the local average treatment effect (LATE) of confinement length. The estimated impact is local in that it is only estimated for the subset of cases for which the instrument - the capacity ratio- induces a change in incarceration lengths (Angrist, Imbens, and Rubin 1996).

The system of equations used in the instrumental variable two stage least squares (2SLS) estimation takes the following form:

$$\widehat{IncarcerationLength} = \pi_1 CapRatio_{ij} + \beta X + e$$

$$FineFeeOrder = \pi_2 \widehat{IncarcerationLength} + \beta X + e$$

$$ProbationLength = \pi_3 \widehat{IncarcerationLength} + \beta X + e$$

In the first stage, we regress incarceration length on the county-level capacity ratio. In that equation, π_1 captures the impact of capacity ratio on sentenced incarceration length. In the second equation, the predicted value of incarceration length obtained in the first-stage equation is used as the focal predictor in lieu of the observed measure. Therefore, the π_2 and π_3 parameters represent the causal effect of incarceration length on each dependent variable, given that the instrumental variable assumptions are satisfied. In the second stage, the standard errors are adjusted by the square of the correlation between incarceration length and the instrument to account for the use of an endogenous variable in the second stage model. The first-stage and second-stage equations include the same set of individual and case controls. Each endogenous variable is logged consistent with the multivariate model discussed above.

Results

Descriptive Analyses

Figure 1 shows the mean fine/fee order by race, with the racial groups ordered by their raw means descending from the highest average. These means are unadjusted for offense mix and severity (among other variables), and represent the simple bivariate relationship between race/ethnicity and monetary sanctions. White defendants are sentenced to $325 of fines and fees on average, with Asian and Hispanic defendants on average facing $327 and $323 in fines and fees, respectively. Native American defendants trail slightly behind Hispanic defendants at $292. Black defendants owe comparatively less at sentencing with mean fine and fee amounts at $272.

Income differs greatly across racial and ethnic categories in Minnesota, so we also adjusted fine and fee amounts for group differences in group-level median income. The income-adjusted amounts are indicated in the striped bars in Figure 1. These can be interpreted as the average amount defendants from each racial group are ordered if the income distributions were equal across groups. The adjusted totals reveal a different picture: relative to income and the ability to pay, Native American and Black defendants are sanctioned most harshly, at $533 and $546 respectively. Conversely, White and Asian defendants pay a lower amount relative to their respective median incomes. These striped bars highlight the increased levels of disparities present when the median level of ability to pay is considered.

In sum, Figure 1 illustrates variation in the sentencing of monetary sanctions by race, and our models below will make similar comparisons, while adjusting for other factors such as offense severity and type, as well as the length of incarceration.

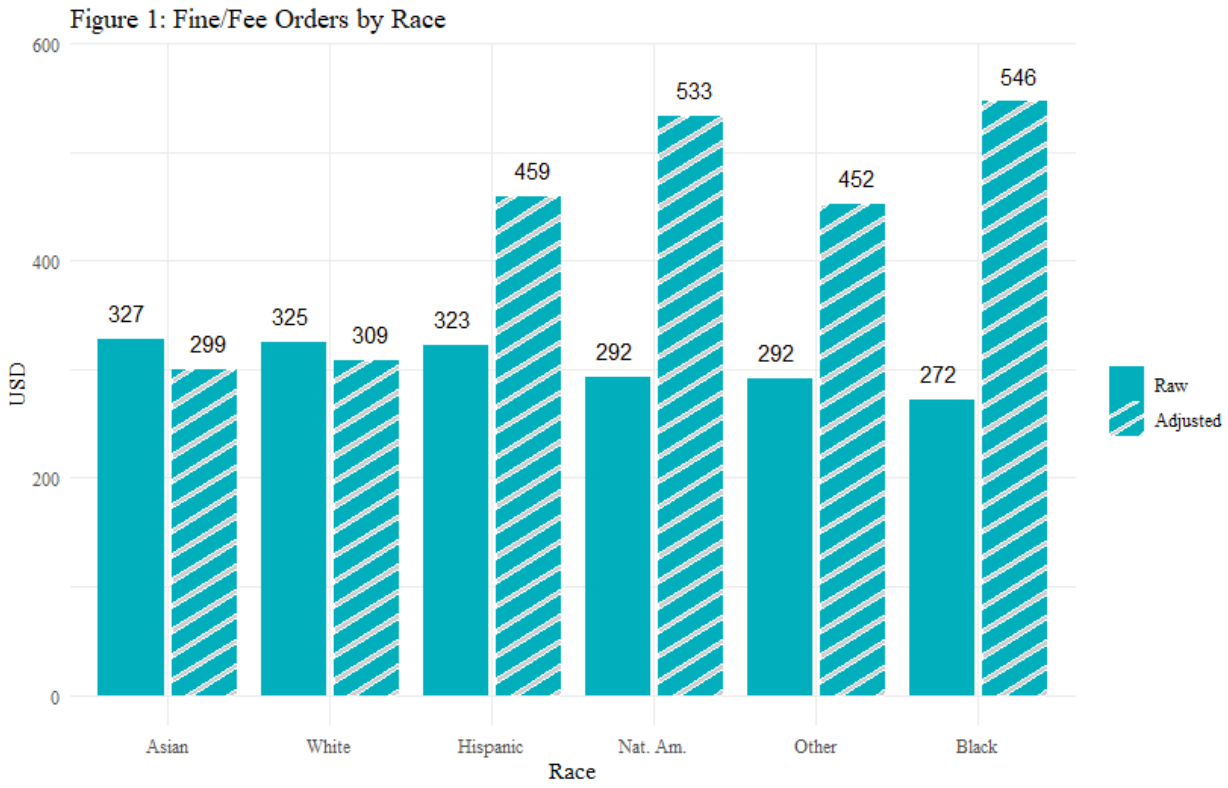


Figure 2 displays the race specific means of our second dependent variable, incarceration length, as measured in expected days served. Native American (50 days) and Black (48 days) defendants receive higher average incarceration lengths as compared to White defendants (26 days). Hispanic (28 days) and Other Race (24 days) defendants, on average, receive similar incarceration sentences to White defendants, and Asian defendants receive five days less on average (21 days). In contrast to the unadjusted race differences in monetary sanctions, Black and Native American defendants receive comparatively more incarceration and lower fines, whereas White defendants receive more fines and less incarceration.

Figure 2: Incarceration Length by Race

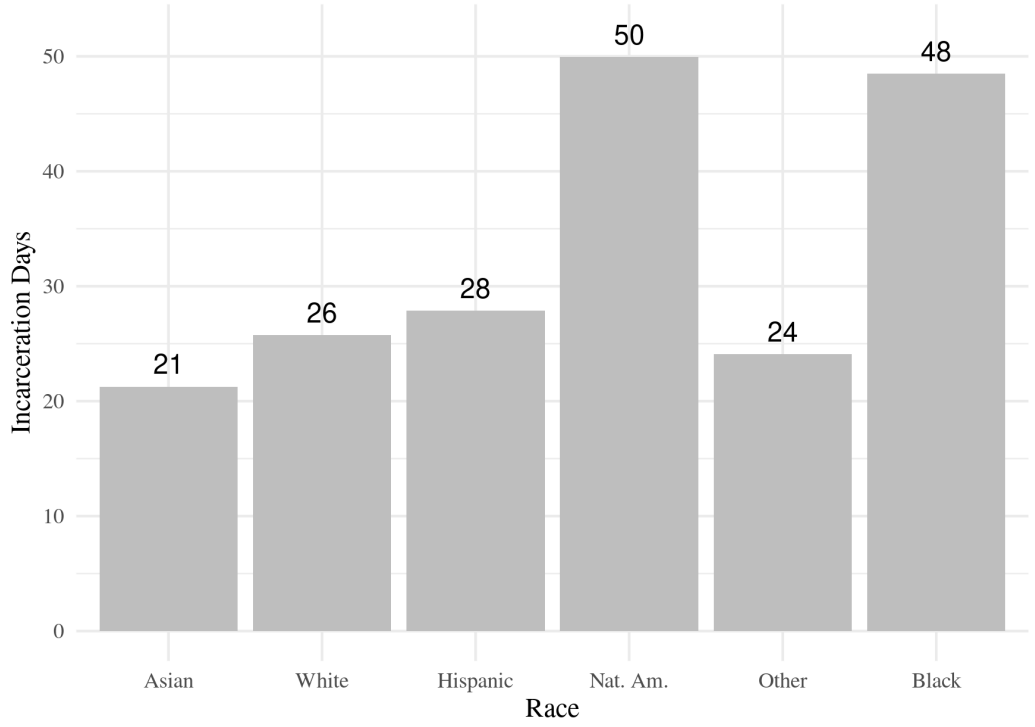
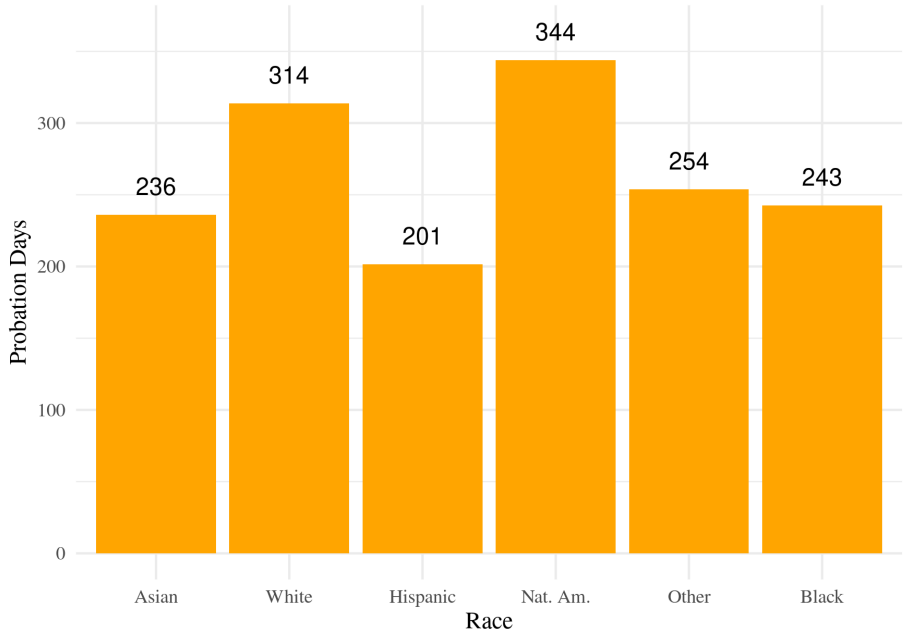


Figure 3 displays the distribution of mean probation sentences by racial group. This graph shows somewhat of a mixture as compared to the confinement and financial penalties graphs. While Native Americans have the highest average probation sentences (344 days), White defendants have the second highest (314 days). In comparison, Black (243 days), Other (254 days), Asian (236 days), and Hispanic (201 days) defendants have lower average probation lengths. In sum, this racialized pattern in probation lengths most closely matches the sentencing of financial punishment, albeit with Native Americans being especially prone to community surveillance in addition to carceral confinement, and Hispanic particularly low in average surveillance lengths.

Figure 3: Probation Length by Race



Multivariate Model of the "Cerberus" of Punishment

Table 2 displays results from a multivariate linear regression of fine and fee amount, incarceration length, and probation length. Estimates in the first column predict the amount of legal financial obligations (LFOs), logged and adjusted for inflation to January 2018 dollars. Black, Native American, and Other Race defendants receive lower fines, net of other factors, as compared to White defendants. Specifically, Black defendants are associated with about 28% lower LFO orders as compared to Whites, and Native American and Other Race receive 8% and 10% lower LFOs respectively. In contrast, Asian and Hispanic defendants receive slightly higher LFOs than White defendants. This provides mixed evidence for H2, as although Hispanic defendants have higher LFO loads relative to White defendants, Black and Native Americans have lower fine and fee amounts. This pattern of results largely parallels the racial patterns in Figure 1, albeit attenuated in the presence of legal and other factors.

The first model also shows the impact of legal factors on LFO amounts. Prior cases, a measure of criminal history, and days credit are associated with lower fines and fees. A case that goes to trial is positively associated with lower LFO orders, with 40% lower fines and fees on average. The presence of a public defender has a relatively large effect, reducing the amounts of fines/fees sentenced, net of other factors, by 61%. This may suggest that judges take into account ability to pay when determining monetary sanctions. Finally, the model includes a measure of offense type. "Non-violent" crimes, such as drug and alcohol related offenses, carry greater financial penalties as compared to violent offenses. Alcohol and DUI offenses carry the greatest average amounts, being 42% greater as compared to violent offenses, all else equal.

The second column in Table 2 shows the estimates in the model predicting incarceration days. Some of the racial effects on incarceration length are directionally opposite their effects in predicting LFOs. Black (.6%), Native American (7%), and Hispanic (1%) defendants receive higher incarceration lengths than White defendants, net of other covariates, which provides support for H1. Although Black, Hispanic, and Native American defendants receive lower fines and fees on average, they are sentenced to higher incarceration lengths, net of other observed measures. These results suggest that the "punishment mix", for similarly situated cases, are patterned by defendant race.

Table 2: Multivariate Model of Minnesota Punishment

	LFO (log)		Incarceration (log)		Probation (log)	
Variables	**β**	**SE**	**B**	**SE**	**β**	**SE**
Extra-Legal						
Race						
Black	-.328***	.003	.006***	.001	-.061***	.006
Hispanic	.049***	.005	.011***	.001	-.412***	.009
Asian	.041***	.007	.002	.002	-.163***	.013
Native American	-.082***	.006	.069***	.002	-.107***	.011
Other Race	-.105***	.009	.001	.003	-.031	.018
Missing Race	-.143***	.003	-.017***	.0009	-1.332***	.006
Male	-.003	.002	.031***	.0007	-.237***	.005
Age (log)	-.052***	.003	.037***	.0009	.463***	.006
Legal						
Prior Cases	-.010***	.0001	.005***	.00004	-.022***	.0003
Public Defender	-.951***	.003	.129***	.0009	2.097***	.006
Credit Days	-.001***	.000009	.0001***	.000009	-.001***	.00002
Trial	-.504***	.117	.558***	.034	-1.864***	.220
Sentence Level						
Felony	.035***	.006	.840***	.002	1.510***	.012
Gross Mis.	.465***	.004	.514***	.001	2.595***	.008
Offense Type						
Drug	.160***	.008	-.130***	.002	-.137***	.015
Alcohol/DUI	.358***	.005	.066***	.001	1.028***	.010
Other Offense	.126***	.005	-.071***	.002	-.501***	.010
Constant	5.605***	.012	-.083***	.001	.839***	.023
Observations	1,371,751		1,371,751		1,371,751	
R^2	.113		.379		.348	
Residual Std. Error	1.296 (*df=1,371,733)*		.379 (*df=1,371,733)*		2.444 (*df=1,371,733)*	
F	10220*** (*df=17, 1,371,733*)		49330*** (*df=17, 1,371,733*)		43140*** (*df=17, 1,371,733*)	

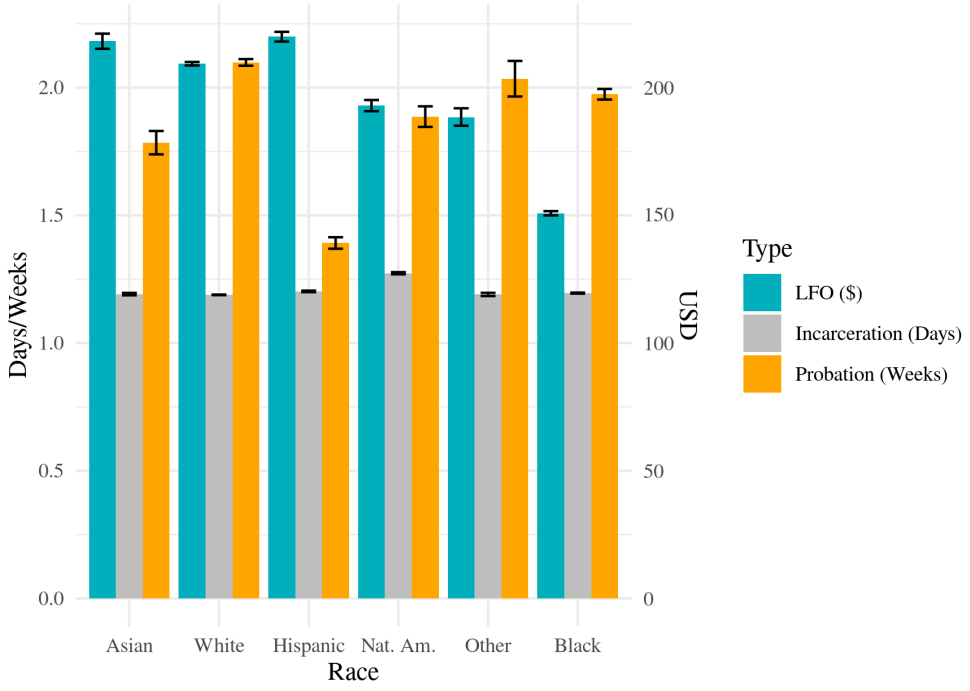
Note: *$p < .05$. **$p < .01$. ***$p < .001$ (two-tailed tests).

The third column displays the estimates on probation length. The racial patterns in probation are primarily directionally opposite to those of confinement. Non-White defendants receive significantly lower probation lengths relative to White defendants, net of other factors. In particular, Hispanic defendants are associated with 33% shorter probation sentences, a relatively large difference. This provides no support for H3, as Black, Hispanic, and Native American defendants have lower probation loads for similarly situated cases.

In sum, the model in Table 2 suggests that Black, Native American, and Other Race defendants receive less punitive sentences in terms of financial punishment and community supervision, but more punitive sentences in terms of carceral punishment, as compared to White

defendants. Overall, these results suggest that more carceral forms of punishment, such as incarceration, are leavied more heavily against non-White defendants, whereas White defendants have the highest probation loads overall. These “punishment packages" for each racial group are estimated in Figure 4. The figure displays the estimated marginal means of each component of the punishment, estimated at mean levels of the continuous covariates and averaged over the levels of categorical covariates. While the racial differences in incarceration are smaller in magnitude, stark differences in both probation length and monetary penalties exist amongst the different racial groups.

Figure 4: Estimated Marginal Punishment Packages



Instrumental Variable Models of the “Cerberus” of Punishment

It is clear from the results in Table 2 and Figure 4 that race is associated with LFOs, incarcations sentences, and probation sentences. Furthermore, these results highlight the variations in racialization present at the sentencing stage. However, if incarceration is associated with either LFOs or probation, then this interrelationship, left unspecified, will bias the residual

race effects in the models. To investigate this possibility, we present the instrumental variable estimates of fine and fee amounts and probation days adjusted for instrumented confinement days.

Table 3: Instrumental Variable Models of LFOs and Probation Length

	LFO (log)		Probation (log)	
Variables	**β**	**SE**	**β**	**SE**
Incarceration Days	-.039***	.004	.009***	.001
Extra-Legal				
Race				
Black	.422***	.079	-.227***	.027
Hispanic	.603***	.067	-.533***	.023
Asian	.269***	.064	-.213***	.021
Native American	.239***	.059	-.177***	.020
Other Race	.136	.082	-.085**	.027
Missing Race	.247***	.046	-1.418***	.015
Male	.422***	.047	-.331***	.016
Age (log)	.344***	.048	.376***	.016
Legal Measures				
Prior Convictions	.020***	.003	-.029***	.001
Public Defender	-.977***	.026	2.103***	.009
Credit Days	.012***	.001	-.004***	.0004
Trial	70.333***	7.100	-17.470***	2.393
Sentence Level				
Felony	13.342***	1.322	-1.422**	.446
Gross Misdemeanor	.729***	.046	2.536***	.015
Offense Type				
Drug	-3.659***	.385	.704***	.130
Alcohol/DUI	-1.894***	.228	1.525***	.077
Other Offense	-2.730***	.286	.128	.096
Constant	6.178***	.118	.713***	.040
IV F	102.6*** (*df=1, 1,371,732*)		102.6*** (*df=1, 1,371,732*)	
Wu-Hausman Test	7112.3*** (*df=1, 1,371,731)*		132.0*** (*df=1, 1,371,731)*	
Observations	1,371,752		1,371,752	
Residual Std. Error	10.84 (*df=1,371,732)*		3.654 (*df=1,371,732)*	
F	143.6*** (*df=18, 1,371,732*)		18220*** (*df=18, 1,371,732*)	

Note: *$p < .05$. **$p < .01$. ***$p < .001$ (two-tailed tests).

The models in Table 3 introduce our instrumental variable estimates of LFO order and probation length. As mentioned above, an F-test of our instrumental variable in the first stage indicates that jail capacity ratios are a sufficiently strong instrument of confinement length ($F = 102.6$, $p < .001$). Additionally, a Wu-Hausman test confirms our expectation that confinement

days are endogenous and correlated with the error term of the LFO equation (F=7112.3, p <.001)[5]. To reiterate the discussion above, the IV estimates remove the simultaneity bias and threat of omitted variables from the estimation of the incarceration length effect, conditional on the IV conditions being satisfied. According to the IV estimates, the causal effect of incarceration days on monetary sanctions is negative, suggestive of a substitution effect, with a 1 day increase in incarceration days resulting in a 4% decrease in monetary sanctions. In the presence of the instrumented incarceration days, the race effects on LFOs are altered. Black defendants, net of incarceration, do not receive lower LFOs than White defendants, suggesting that the negative coefficient observed in Table 2 is due to the "trade-off" between incarceration and monetary sanctions.Thus, Black defendants actually receive 53% *higher* fines and fees on average, after incarceration is accounted for. The Asian and Hispanic associations are still statistically significant and positive, and the Native American coefficient flips to positive, suggesting that, controlling for incarceration, these minority racial groups are punished more harshly in terms of financial punishment as compared to White defendants. The race effects are relatively large: Hispanic and Asian defendants are sentenced to 83% and 31% higher LFOs as compared to White defendants, whereas Native American are ordered to 27% more respectively. These findings suggest that part of the racial disparities in financial punishment above were driven by racial differences in incarceration length, and mask the race disparities in monetary sanctions. Further, they illustrate that the racial disparities are not a mere economic trade-off between different axes of punishment, as would be the case if the race effects went to zero in the presence of confinement length.

[5] The Wu-Hausman test is an F-test of the second stage model but including the residuals from the first stage model. If the residuals from the first stage model (i.e, the variation in incarceration days hypothesized to be correlated with the error term) have a significant impact, that is evidence of endogeneity. The null hypothesis is that the increase in regression sum of squares in the second stage model upon introduction of the first stage residuals is equal to 0.

In terms of probation, instrumented incarceration length is positively associated, with a 1% increase in confinement length associated with a 1% increase in probation days. This is suggestive of a "piling on" effect, as is contrasted with the substitution present between fine/fees and incarceration. In terms of the extra-legal race effects, non-White racial groups are significantly associated with lower levels of probation, net of the incarceration sentence. This finding, in the context of the incarceration model above, suggests that while White defendants receive lower incarceration and more probation for similarly situated cases, non-White defendants receive more incarceration and less probation on average. Further, the race effects here are not fully explained by controlling for incarceration length, suggesting that the lower probation amounts for non-Whites are not solely due to their higher incarceration loads.

Discussion and Conclusion

Although incarceration has been a principal focus of the sociological study of punishment, relatively less attention has considered the broader package that constitutes American punishment. In this paper, we have examined how the racialization of punishment may depend on the particular facet of the punishment matrix being analyzed. Specifically, we find that the "mix" of punishment that defendants receive is patterned by race. Black and Native American defendants, descriptively, generally receive lower fine and fee amounts, but this is offset by longer incarceration sentences.

However, when we appropriately account for incarceration in the sentencing of both probation and monetary sanctions, non-White defendants tend to receive *higher* monetary sanctions. While this is largely consistent with previous work in other social contexts that finds positive race effects, particularly in terms of incarceration, if we were not to appropriately

specify incarceration as part of the monetary sanctions model our conclusions would be mistaken: that minority defendants receive a "break" when it comes to monetary sanctions. To the contrary, our instrumental variable models - which appropriately control for incarceration - show that, compared to White defendants, minority defendants receive higher LFOs once their incarceration sentence is taken into account. When extant levels of racial variation in defendant ability to pay are considered (see Figure 1), the race effects in the model likely *understate* the differential impact monetary sanctions have across racial groups.

The incarceration pattern is reversed for probation: White defendants have the highest probation loads on average, and this is not fully explained by their lower incarceration lengths. Thus, it appears that probation is racialized in a different way as compared to incarceration and monetary sanctions. Importantly, these models demonstrate that the racial disparities in monetary sanctions and probation *are not fully explainable* by a pure economic "trade-off" between different forms of punishment. This would be the case if incarceration length fully explained the race effects present in the model. However, we observe statistically significant, robust race impacts, net of incarceration loads on both monetary sanction and probation sentences. This highlights how race plays a significant part of the sentencing decision, *above and beyond its impact on the focal punishment, incarceration.* In other words, racial disparities in monetary sanction and probation are not mere downstream consequences of extant disparities in incarceration, but rather are terrain amenable to racial disparities of their own.

In total, our analysis provides further evidence of the racialized character of modern punishment in America. Specifically, the results here are consistent with the research on extra-legal race effects on incarceration (e.g., Spohn 2000), and our models extend these effects to both probation and monetary sanctions. Further, our results highlight the persistent and

ubiquitous non-White sentencing disparities across incarceration and monetary sanctions. Not only are non-White defendants sentenced to longer incarceration sentences, but are also sentenced to higher fine/fee amounts as compared to White defendants when controlling for the already existing disparities in incarceration. This is further evidence suggesting that the criminal legal system continues to play a prominent role as a racialized social control institution (e.g., Waquant 2009). However, these racial impacts are not solely the purview of Blackness, as we document consistent extra-legal effects for both Hispanic and Native American defendants in terms of both incarceration and monetary sanctions. In contrast to the racialized patterns in incarceration and monetary sanctions, we observe lower probation amounts for non-White defendants, even above and beyond the impact of incarceration lengths. This suggests that Whiteness is associated with "less severe" forms of punishment, and further illustrates that the "packaging" punishment is racialized in complex ways.

Additionally, the analysis here indicates that not only has crime become racially typified in American culture (e.g., Quillian and Pager 2001), but that punishment, at least in the decisions of courtroom judges, has a racially typified character as well. Even in an era of sentencing guidelines and the standardization of sentences (Ulmer 1997), race continues to have an impact on the modality and severity a defendant is likely to encounter. This has severe implications considering the considerable "collateral consequences" (Uggen and Stewart 2014) one experiences as part of the criminal justice system, and the role these disparities play in the reproduction of American racial stratification (Uggen and Wakefield 2010).

Our analysis here is not meant to replace existing models of racial effects on punishment, but rather to extend them and illustrate the interrelationships between varied modes of punishment and capture a more complete picture of the "cerberus" of punishment. This suggests

that studies of race and incarceration would benefit from considering monetary sanctions and other forms of punishment in tandem. Similarly, studies of monetary sanctions may be incomplete without considering their relationship to incarceration sentences. Such analyses are consistent with how punishment is experienced by defendants, who often receive a package of monetary and non-monetary sanctions.

Further, our analysis not only highlights the packaging of punishment by race, but how the estimation of racial and other effects on punishment can be misleading if the analyses do not take into account the other concurrent forms of punishment that are levied at sentencing. We find a negative causal effect of confinement length on monetary sanctions, and a positive incarceration effect on probation lengths. Controlling for the incarceration effect alters the magnitude, *and the direction*, of many of the race indicators. Therefore, by considering any facet of punishment in isolation risks missing out on how that punishment fits into the overall package of punishment, yielding biased or distorted estimates of how race and other factors affect punishment. Considering these parts of punishment in tandem increases our ability to examine the ways in which these sentencing outcomes are shaped by race and other factors. In sum, our analysis suggests that researchers should consider multiple facets of the punishment nexus to get a more complete, and arguably more accurate, portrayal of punishment.

A limitation to the current study is that we only consider the confinement, probation, and monetary components of punishment, and do not consider other aspects such as other court-ordered requirements such as mental health treatment or drug testing. While some of these facets may remain charge type specific, they could be collinear with key variables and the components of punishment, and therefore confound our race coefficients. Future research should expand the punishment outcomes to gain a more complete and resolute picture of the

“punishment package” amongst different racial groups. Moreover, the SCAO data are limited in terms of measures at the defendant level, missing measures such as income and educational status. These measures could be further axes of stratification within the sentencing stage of punishment, and are highly correlated with race (Manduca 2018; Kao and Thompson 2003). Minnesota also may be an outlier in terms of the use of LFOs and other forms of punishment, as other research shows Minnesota gives comparatively less fines and fees than many other states (Harris et. al. 2017).

The relative scarcity of research analyzing multiple components of punishment may be due partially to the empirical modeling challenges discussed above, as well as access to complete and robust individual-level sentencing data. Notwithstanding these caveats, however, our analysis provides a foundation for future work to consider the nexus of race and the myriad forms of punishment and illustrates that punishment should be modeled as it is often experienced: as a constitutive package of different forms and intensities of punishment.

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